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## APPENDIX A. DIFFERENTIATION FORMULA FOR VARIABLE-LIMIT INTEGRALS(LEIBNIZ RULE)

$\Omega(t)$  denotes a time-dependent bounded domain, and at time  $t^*$ ,  $\Omega(t)$  is fixed as  $\Omega_t^*$ , i.e.,  $\Omega_t^* = \Omega(t^*)$ . Let  $\frac{d}{dt}(\cdot)$  represent the material derivative (substantial derivative), and denote  $\mathbf{u} = \frac{dx}{dt}$ . The differentiation formula for integrals with variable limits (also known as the Leibniz rule) is given by: